

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

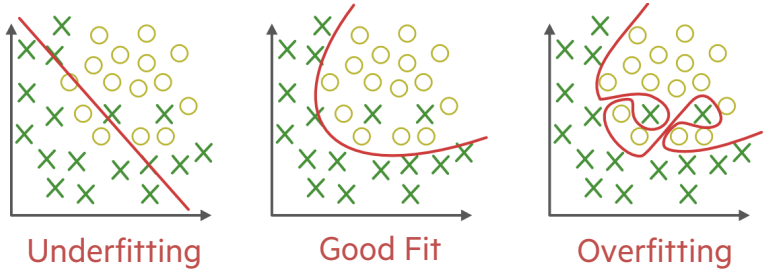
- Classwork 5
  - Due **tomorrow, 11/8 @5pm**
- Homework 3
  - Due next Friday, 11/15 @5pm
- Final Exam
  - Thursday 12/12 @2pm

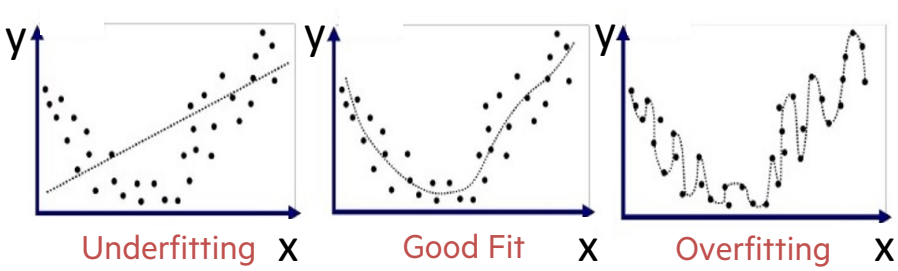
This Lecture

- Perceptron
  - Neuron vs Perceptron
  - Linear Perceptron
  - Learning Perceptron

Last Lecture

- Overfitting
  - High accuracy on **training** data
  - Poor performance on new, unseen data (**validation/test** sets)





- Overfitting in NBC
  - Limited Training Data: $P(x_i|y)$ less accurate
  - Zero Probability Problem: $P(x_i|y) = 0$ for unseen words

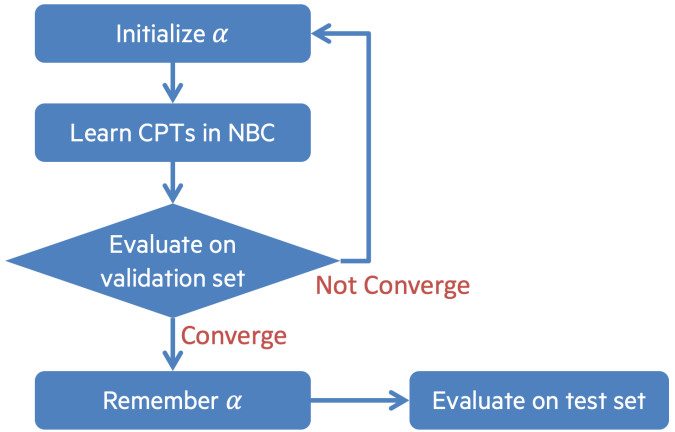
Last Lecture

- Smoothing in NBC

$$P(x_i|Y = spam) = \frac{N_{x_i|spam} + \alpha}{N_{spam} + \alpha N_{vocab}}$$

$$P(x_i|Y = ham) = \frac{N_{x_i|ham} + \alpha}{N_{ham} + \alpha N_{vocab}}$$

- Hyperparameter Tuning
  - Parameter vs. Hyperparameter
  - Hyperparameter Tuning/Searching



Last Lecture

- Evaluation
  - Classification
    - Accuracy
    - Precision
    - Recall
    - F1 Score
  - Regression
    - MSE
    - RMSE
    - MAE
    - R^2

Example: Naïve_Bayes.ipynb

“Weighting” Features

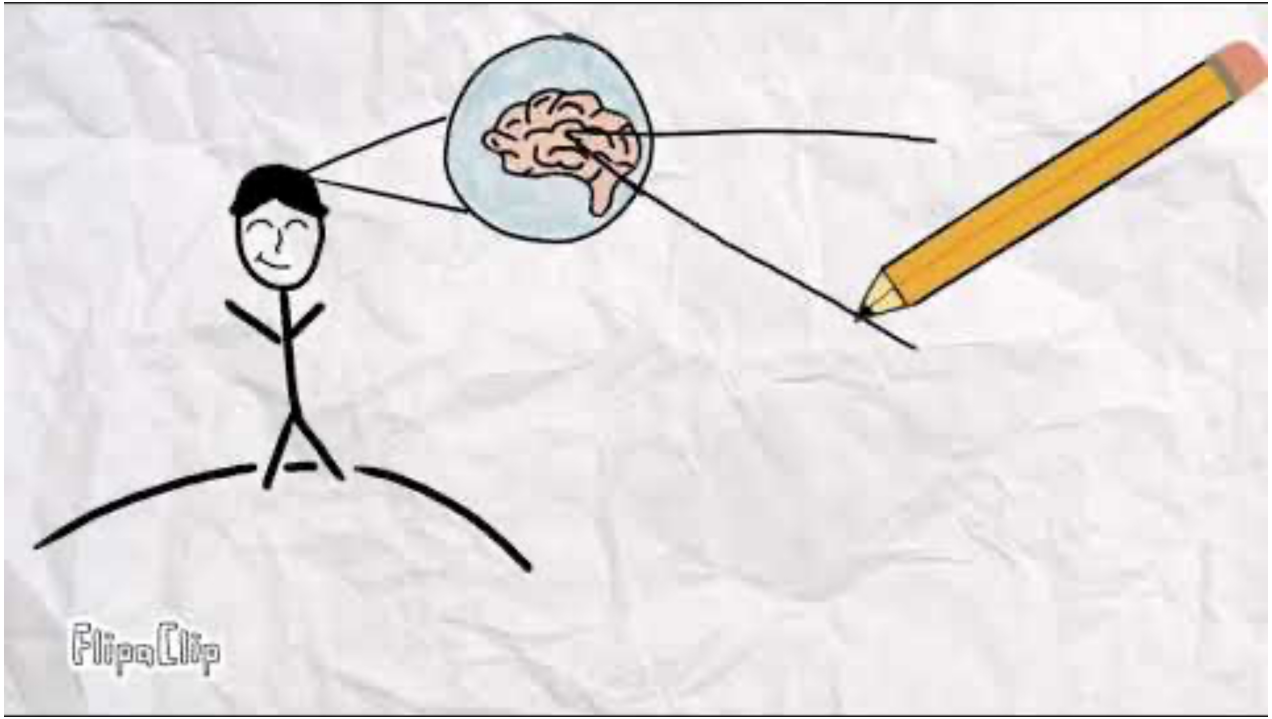
- NBC for Spam Detection
  - Other informative features?
    - Sender’s email domain
    - Whether or not sender is in your contact
  - How to add these feature in NBC?
    - Additional CPTs

What are the limitations?

index	Word 1	Word 2	…
1	0	1	
2	1	1	
…			

email	contact
lmu.edu	Yes
xa3h.com	No
…	

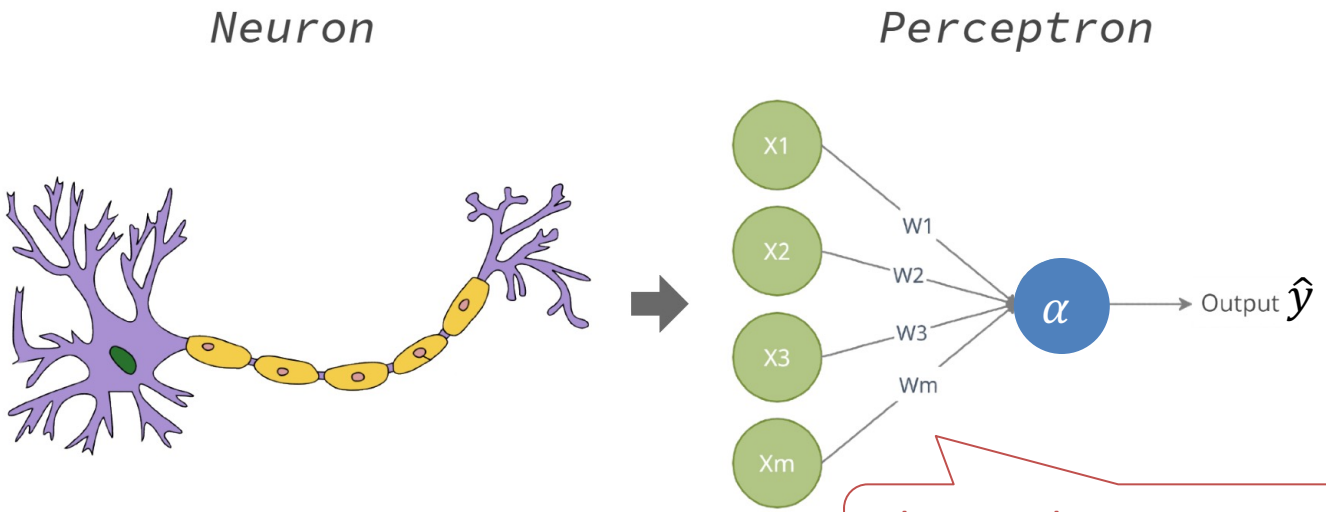
Neurons and Neural Networks



Video Link: https://www.youtube.com/watch?v=AthtkYkNcOM

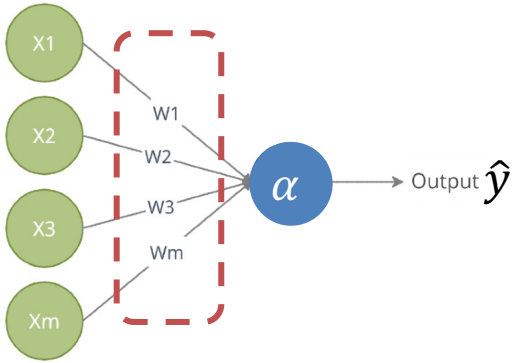
Perceptron

- Represent “weighted” feature for classification



Perceptron's Parameters

- Weight Vector w
  - Each input feature x_i is **weighted** by a scalar w_i for how much the x_i contributes to classification.
    - Can be **positive** (indicative) or **negative** (non-indicative)
    - Can be **large** or **small**



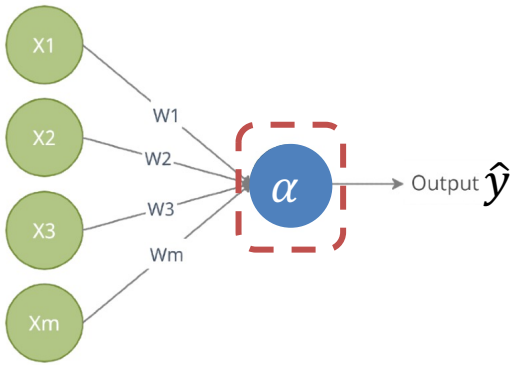
Perceptron's Parameter

- Activation Function $\alpha = \text{activation}_{\boldsymbol{w}}(\boldsymbol{x})$
  - Takes the weight vector $\boldsymbol{w}$ and input feature vector $\boldsymbol{x}$ and produces some activation α
  - For linear perceptron, $\alpha = \sum_i w_i \cdot x_i$ (a dot product of $\boldsymbol{w}$ and $\boldsymbol{x}$)

$$\vec{A}^T = \begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix} \qquad \vec{B} = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix}$$

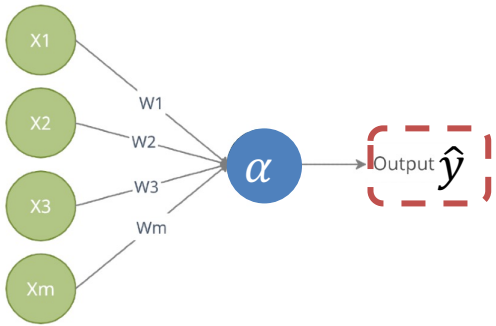
$$\begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix} = A_1B_1 + A_2B_2 + A_3B_3 = \vec{A}.\vec{B}$$

Dot Product



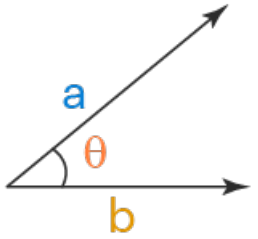
Perceptron's Parameter

- Decision Rule $\hat{y} = \text{decision}(\alpha)$
  - Assigns a class $\hat{y}$ based on the activation α for a given weighted input.
  - Binary decision rule
    - $|Y| = 2$ (e.g., Y={spam, ham})



Binary Decision Rule

- $\boldsymbol{w}$ and $\boldsymbol{x}$ are n-dimensional vectors
  - $\boldsymbol{w} =< w_1, w_2, \ldots, w_n >$ and $\boldsymbol{x} =< x_1, x_2, \ldots, x_n >$
- $\alpha = \boldsymbol{w} \cdot \boldsymbol{x} = \sum_i w_i \cdot x_i$ is a scalar value
  - $\alpha > 0$: $\boldsymbol{w}$ and $\boldsymbol{x}$ in the same direction
  - $\alpha < 0$: $\boldsymbol{w}$ and $\boldsymbol{x}$ in the opposite direction
- Binary Perceptron Decision Rule
  - $\hat{y} = (\text{activation}_{\boldsymbol{w}}(\boldsymbol{x}) > 0)?\, 0 : 1$



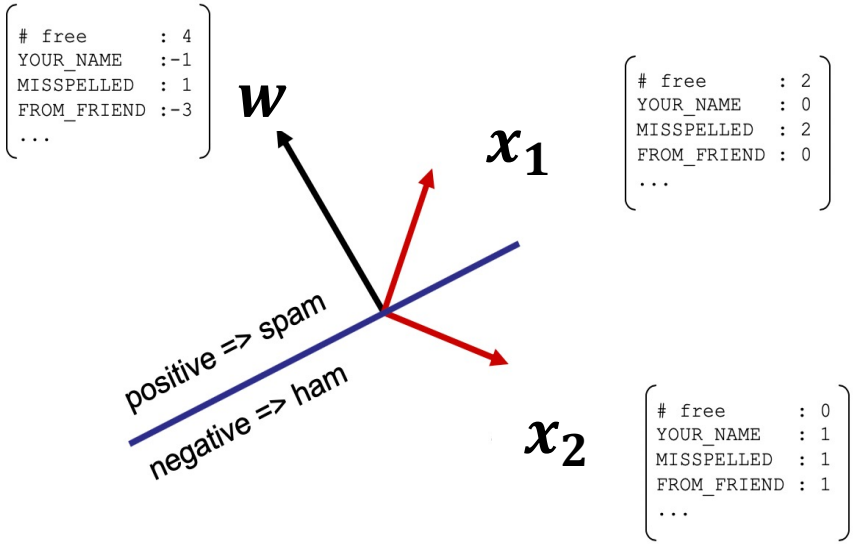
a . b = |a||b| cos θ

if true

if false

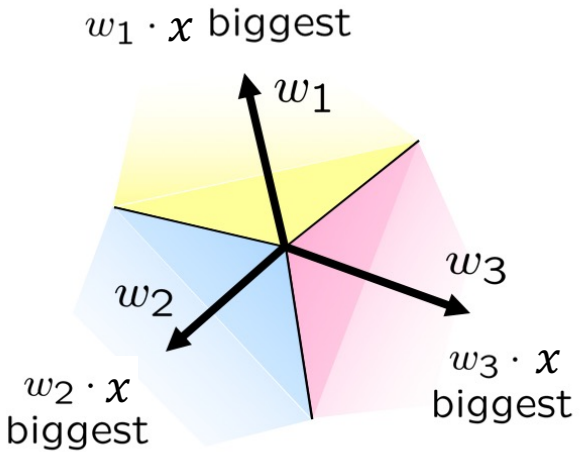
Example

- Compute the activation and assign emails $\boldsymbol{x_1}$ and $\boldsymbol{x_2}$ to the binary spam/ham class.
  - $\alpha_1 = 8 + 0 + 2 + 0 = 10$
    - Spam
  - $\alpha_2 = 0 - 1 + 1 - 3 = -3$
    - Ham



Multi-Class Perceptron

- How about multi-class classification?
  - Binary decision rule (+/-) no longer works
- Multi-Class Perceptron
  - Each class y has its own weight vector $\boldsymbol{w_y}$ that must be record-kept separately
  - Each input has a class-specific activation
    - $a_y = \text{activation}_{\mathbf{w_y}}(\boldsymbol{x})$



Multi-Class Perceptron

- Decision Rule
  - Which class is $\boldsymbol{x}$ most like? Which y has biggest α_y?
  - $\hat{y} = \text{argmax}_y \alpha_y = \text{argmax}\, \boldsymbol{x} \cdot \boldsymbol{w_y}$
- Binary vs. Multiclass Perceptron
  - Binary: Yes/No
  - Multiclass: Most likely

Example

- 3 features, and 3 classes {0,1,2}
  - $\boldsymbol{w_0} = < 0,1,2 >, \boldsymbol{w_1} =< -1, -2,0 >, \boldsymbol{w_2} =< 2,0,2 >$
  - $\boldsymbol{x} =< 1,3,2 >$
- Which class would the perceptron assign to the input $\boldsymbol{x}$?
  - $\alpha_0 = 0 + 3 + 4 = 7$ $\hat{y} = 0$
  - $\alpha_1 = -1 - 6 + 0 = -7$
  - $\alpha_2 = 2 + 0 + 4 = 6$

Remaining…

CW5 & HW3